Graduation Project Management System (GPMS)

Requirements Categorized by Major User Roles

Scope: Functional Requirements Specification **Due Date:** November 10, 2026

General System & Security Framework (All Roles)

- **Authentication & SSO:** Supports secure university Single Sign-On (OAuth2 / SAML) or standard email/password authentication.
- **Profile Management:** Maintains detailed user profiles displaying academic departments, skills, interest areas, contact info, and availability.
- **Real-Time Notifications:** Delivers in-app and email alerts for deadline reminders, submission feedback, approval decisions, and meeting invites.

1. Students

- **Team Formation:** Can create, join, and invite peers to teams, adhering to strict team size limitations dictated by the project scope.
- **Proposal Management:** Can request available topics provided by the faculty catalog, or draft, edit, and submit custom project proposals for review.
- **Task Board:** Utilizes task tracking to create, assign, and update sub-tasks with deadlines, priorities, and status tags (To Do, In Progress, Review, Done).
- **File Submission Portal:** Performs version-controlled file uploads for documentation (PDF, DOCX) and project assets (ZIP, code repositories).
- **Team Collaboration:** Accesses a private discussion board and project feed per team for sharing notes, links, and supervisor announcements.
- **Meeting Coordination:** Synchronizes with integrated calendars (Google Calendar / Outlook) for setting up supervisor-team weekly catch-ups.

2. Supervisors / Advisors

- **Advisor Matching:** Matched automatically with student teams based on project specifics and supervisor availability.
- **Proposal Review:** Participates in the multi-stage review pipeline by evaluating and reviewing draft project proposals.
- **Project Supervision & Workflow:** Reviews student submissions during intermediate stages (such as Supervisor Review) and provides structured feedback.
- **Meeting Coordination:** Coordinates weekly catch-ups and meetings via calendar integrations (Google Calendar / Outlook).
- **External/Committee Evaluation:** Collaborates with external examiners/committee members during evaluations (note: external examiners also leverage evaluation tools/rubrics).

3. Admins (Department Coordinators)

- **Role-Based Access Control (RBAC):** Oversees distinct permissions and dashboard configurations across Students, Supervisors, Admins, and External Examiners.
- **Proposal Workflow Administration:** Manages the multi-stage review pipeline (Draft -> Under Supervisor Review -> Department review -> Accepted / Revision Required / Rejected).
- **Timeline & Schedule Management:** Sets university-wide deadlines for milestones such as Project Proposal, SRS Document, Prototype, Final Thesis, and Defense.
- **Automated Penalty Enforcement:** Enforces business rules that automatically mark submissions past the deadline as 'Late' or lock submissions entirely.
- **Analytics & Dashboards:** Utilizes high-level administrative dashboards to track active projects, supervisor workloads, submission completion rates, and department performance.
- **Exportable Reporting:** Exports student grades, team rosters, and evaluation summaries to Excel, PDF, or CSV for official university archiving.
- **Public Repository & Archiving:** Maintains the searchable digital library of past approved projects, abstracts, tech stacks, and showcase work for future students.